

2D Imager Barcode Scanner

- MS852Plus -



User's Manual

Version 1.0

Change Log

Date	Change Description	Version
2019/6/14	first published version	1.0

Preface

About This Manual

Thank you for purchasing the unitech product.

This manual explains how to install, operate and maintain our product.

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Regulatory Compliance Statements



FCC Warning Statement

This device has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference with radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference with radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

1. This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.
2. This device complies with FCC RF radiation exposure limits set forth for an uncontrolled environment. To maintain compliance with FCC RF exposure requirements, avoid direct contact to the transmitting antenna during transmitting.
3. Any changes or modifications (including the antennas) made to this device that are not expressly approved by the manufacturer may void the user's authority to operate the equipment.

Operation on the 5.15 - 5.25GHz frequency band is restricted to indoor use only. The FCC requires indoor use for the 5.15-5.25GHz band to reduce the potential for harmful interference to co-channel Mobile Satellite Systems. Therefore, it will only transmit on the 5.25-5.35 GHz, 5.47-5.725 GHz and 5.725 - 5.850 GHz band when associated with an access point (AP).

FCC Label Statement

This device complies with part 15 of the FCC rules. Operation is subject to the following two conditions:

1. This device may not cause harmful interference.
2. This device must accept any interference received, including interference that may cause undesired operation.

RF Radiation Exposure Statement

For body contact during operation, this device has been tested and meets FCC RF exposure guidelines when used with an accessory that contains no metal and that positions the handset a minimum of 1.5 cm from the body. Use of other accessories may not ensure compliance with FCC RF exposure guidelines.

Canadian Compliance Statement

This Class B Digital apparatus meets all requirements of the Canadian Interference-Causing Equipment Regulations.

Cet appareil numérique de la classe B respecte les exigences du Règlement sur le matériel brouilleur du Canada.

European Conformity Statement

unitech Electronics co., Ltd herewith declares that the unitech product is in compliance with the essential requirements and all other provisions of the RED 2014/53/EU directive, the EMC 2014/30/EU directive and the Low Voltage 2014/35/EU directive.

The declaration of conformity is available for download at :

<https://portal.unitech.eu/public/Safetyregulatorystatement>

CE RF Exposure Compliance

This device meets EU requirements (2014/53/EU) on the limitation of exposure of the general public to electromagnetic fields by way of health protection.

For body-worn operation, this device has been tested and meets the ICNIRP guidelines and the European Standard EN 62209-2, for use with dedicated accessories, SAR is measured with this device at a separation of 0.5 cm to the body, while transmitting at the highest certified output power level in all frequency bands of this device. Use of other accessories which contain metals may not ensure compliance with ICNIRP exposure guidelines.

CE Mark Warning



This equipment complies with the requirements of Directive 2014/53/EU of the European Parliament and Commission from 24 May, 2014 governing Radio and Telecommunications Equipment and mutual recognition of conformity.

RoHS Statement



This device conforms to RoHS (Restriction of Hazardous Substances) European Union regulations that set maximum concentration limits on hazardous materials used in electrical and electronic equipment.

Waste electrical and electronic equipment (WEEE)



unitech has set up a policy and process to meet the EU directive 2002/96/EC and update 2003/108/EC concerning electronic waste disposal.

For more detailed information of the electronic waste disposal of the products you have purchased from unitech directly or via unitech's resellers, you shall either contact your local supplier or visit us at :

<https://portal.unitech.eu/public/WEEE>

Taiwan NCC Warning Statement

低功率電波輻射性電機管理辦法

第十二條：經型式認證合格之低功率射頻電機，非經許可，公司、商號或使用者均不得擅自變更頻率、加大功率或變更原設計之特性及功能。

第十四條：低功率射頻電機之使用不得影響飛航安全及干擾合法通信；經發現有干擾現象時，應立即停用，並改善至無干擾時方得繼續使用。

前項合法通信，指依電信法規定作業之無線電通信。

低功率射頻電機需忍受合法通信或工業、科學及醫療用電波輻射性電機設備之干擾。

注意事項：

1. 使用過度恐傷害視力。
2. 使用30分鐘請休息10分鐘；2歲以下幼兒不看螢幕，2歲以上每天看螢幕不要超過1小時。
3. 減少電磁波影響，請妥適使用。

Laser Information

The unitech product is certified in the U.S. to conform to the requirements of DHHS/CDRH 21CFR Subchapter J and to the requirements of IEC 825-1. Class II and Class 2 products are not considered to be hazardous. The unitech product contains internally a Visible Laser Diode (VLD) whose emissions do not exceed the maximum limits as set forth in the above regulations. The scanner is designed so that there is no human access to harmful laser light during normal operation, user maintenance or prescribed service operations.

The laser safety warning label required by the DHHS/IEC for the unitech product's optional laser scanner module is located on the memory compartment cover, on the back of the unit.

* Laser information only applies to the products with laser components.

CAUTION! Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous laser light. Use of optical instruments with the scanner, including binoculars, microscopes, and magnifying glasses, with will increase eye damage. This does not include eyeglasses worn by the user.

LED Information

The unitech product contains LED indicator(s) or LED ring whose luminance is not harmful to human eyes during normal operation, user maintenance or prescribed service operations.

*LED information only applies to the products with LED components.

Battery Notice

1. To guarantee optimal performance, it is recommended that rechargeable batteries be replaced every year, or after 500 charging cycles are completed. It is normal for the battery to balloon or expand after one year or 500 cycles. Although it does not cause damage, it cannot be used again and must be disposed of according to the location's safe battery disposal procedures.
2. If a battery performance decreases more than 20%, the battery is at the end of its life cycle. Stop use and ensure the battery is disposed of properly.
3. The length of time that a battery lasts depends on the battery type and how the device is used. Conserve the battery life by doing the following:
 - Avoid fully uncharging the battery because this places additional strain on it. Several partial uncharges with frequent charges are better than a fully uncharged battery. Charging a partially charged battery does not cause harm to the unit.
 - Keep the battery cool. Avoid hot vehicles. For prolonged storage, keep the battery at a 40% charge level.
 - Do not leave the battery uncharged and unused for an extended period of time, the battery will wear out and the longevity of the battery will be at least half of one with frequent charges.
4. Protect battery life by not over or under charging the battery.
5. Please do not leave battery unused for long time without charging it. Despite unitech's safety precautions, the battery pack may begin to change shape. If so, stop using it immediately. Please check to see if you are using a proper power adapter to charge the battery or contact your service provider for service.
6. If you cannot charge the battery after it has been idle for an extended period of time and it begins to heat up, please do not try to charge it. It may not be functional anymore.
7. Please only use the original battery from unitech. Using a third party battery can damage our products. Please note that when such damage occurs, it is not covered by your warranty.

CAUTION!

- RISK OF EXPLOSION IF BATTERY IS REPLACED INCORRECTLY.
DISPOSE OF USED BATTERIES ACCORDING TO THE INSTRUCTIONS.
- 如果更換不正確之電池行事會有爆炸的風險
請依製造商說明書處理用過之電池
- 如果更換不正確之電池行事會有爆炸的風險
請依製造商說明書處理用過之電池

Battery charge notice

It is important to consider temperature when the battery pack is charging. Charging is most efficient at normal room temperature or in a slightly cooler environment. It is essential that batteries are charged within the stated range of 0°C to 40°C. Charging batteries outside of the specified range could damage the batteries and shorten their life cycle.

CAUTION! Do not charge batteries at a temperature lower than 0°C. This will and make the batteries unstable and dangerous. Please use a battery temperature detecting device for a charger to ensure a safe charging temperature range.

CAUTION! To ensure the unit working properly, please keep all connectors away from the contaminants staying inside of them such as dust, grease, mud, and water. The negligence may cause the unit with no communication, short circuited, overheated and so on.

CAUTION! If the connector is damaged, please ensure the connector is being fully repaired before use the unit to avoid causing short circuited.

Storage and safety notice

Although charged batteries may be left unused for several months, their capacity may be depleted due to build up of internal resistance. If this happens, they will require recharging prior to use. Batteries may be stored at temperatures between -20°C to 60°C, however they may deplete more rapidly at higher temperatures. It is recommended to store batteries at room temperature.

** The message above only applies to the usage of the removable batteries.
For the products with non-removable batteries / without batteries, please refer to the specification of each product.*

Product Operation and Storage Notice

The unitech product has applicable operation and storage temperature conditions. Please follow the limitation of suggested temperature conditions to avoid failure, damage or malfunction.

**For applicable temperature conditions, please refer to the specification of each product.*

Adapter Notice

1. Please do not leave the power adapter in the socket when it is not connected to your unitech product for charging.
2. Please remove the power adapter when the battery is fully recharged.
3. The bundled power adapter that comes with your unitech product is not meant to be used outdoors. An adapter exposed to water or rain, or a very humid environment can cause damage to both the adapter and the product.
4. Please only use the bundled power adapter or same specification of adapter to charge your unitech product. Using the wrong power adapter can damage your unitech product.

** The message above only applies to the product connected to the adapter.
For the products without using the adapters, please refer to the specification of each product.*

Hearing Damage Warning

Zx.3 Warning

The warning shall be placed on the equipment, or on the packaging, or in the instruction manual and shall consist of the following:

- the symbol of Figure 1 with a minimum height of 5 mm; and
- the following wording, or similar :

To prevent possible hearing damage, do not listen at high volume levels for long periods.



Figure 1 – Warning label (IEC 60417-6044)

Alternatively, the entire warning may be given through the equipment display during use, when the user is asked to acknowledge activation of the higher level.

Worldwide Support

unitech's professional support team is available to quickly answer questions or assist with technical-related issues. Should an equipment problem occur, please contact the nearest unitech regional service representative.

For complete contact information please visit the Web sites listed below:

Taipei, Taiwan – Headquarters Tel: +886-2-89121122 E-mail: info@hq.ute.com Address: 5F, No. 136, Lane 235, Baoqiao Road, Xindian District, New Taipei City 231, Taiwan (R.O.C.) Website: http://www.ute.com	Europe Tel: +31-13-4609292 E-mail: info@eu.ute.com Address: Kapitein Hatterasstraat 19, 5015 BB, Tilburg, the Netherlands Website: http://eu.ute.com
China Tel: +86-59-2310-9966 E-mail: info@cn.ute.com Address: Room401C, 4F, RIHUA International Mansion, Xinfeng 3rd Road, Huoju Hi-tech District, Xiamen, Fujan , China Website: http://cn.ute.com	Japan Tel: +81-3-35232766 E-mail: info@jp.ute.com Address: Kayabacho Nagaoka Building 8F.,1-5-19 Shinkawa, Chuo-Ku, Tokyo, 104-0033, Japan Website: http://jp.ute.com
Asia & Pacific / Middle East Tel: +886-2-27911556 E-mail: info@apac.ute.com info@india.ute.com info@mideast.ute.com Address: 4F., No. 236, ShinHu 2nd Rd., NeiHu Chiu, 114, Taipei,Taiwan Website: http://apac.ute.com / http://mideast.ute.com	Latin America Tel: +52-55-5171-0528 E-mail: info@latin.ute.com Address: 17171 Park Row, Suite 210 Houston, TX 77084USA (Rep.) Website: http://latin.ute.com
North America Tel: +1-714-8926400 E-mail: info@us.ute.com / info@can.ute.com Address: 6182 Katella Ave, Cypress, CA 90630, USA Website: http://us.ute.com	Please scan QR Code to visit us : 

Warranty Policy

The items covered under the unitech Limited Warranty are free from defects during normal use.

The warranty period is varied from each country. Please consult with your supplier or unitech local office for actual length of warranty period to your purchased product.

Warranty becomes void if equipment is modified, improperly installed or used, damaged by accident or neglect, or if any parts are improperly installed or replaced by the user.

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Chapter 1 - Overview

1.1 Package

Please make sure the following contents are in the MS852Plus gift box.
If something is missing or damaged, please contact your unitech representative.

The standard package contents:

- MS852Plus 2D Imager Barcode Scanner
- Cable
- Quick Start Guide
- Regulatory Compliance Statements

NOTE: The barcode with an asterisk (*) which appears in the following chapters indicates that it is the default option for the corresponding setting.

1.2 Product Detail



1.3 Specifications

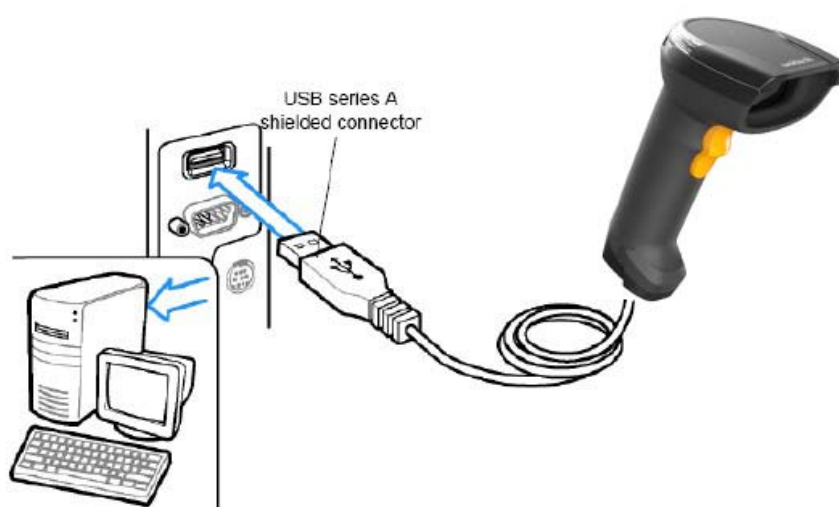
Optical & Performance	
1D or 2D	2D
Sensor	1280 X 800 global shutter
Aiming Element	Red laser
Illumination	White LED (exempt risk group)
Imager Field of View	100,000 Lux (Sunlight)
Skew Angle	± 60°
Pitch Angle Sensor	± 60°
Roll Angle	360°
Optical Resolution	1D 3mil/ 0.075mm
Printing Contrast Scale	20% minimum print contrast ratio
Depth of Field (DOF PCS=80%)	SYMBOLGY / X-DIM TYPICAL RANGE* Near Far 13 mil UPC 1.34 in / 3.4 cm 22.5 in / 57.3 cm 5 mil Code 39 2.76 in / 7 cm 11.85 in / 30.1 cm 10 mil Code 39 1.14 in / 2.9 cm 20.3 in / 51.7 cm 20 mil Code 39 1.38 in / 3.5 cm 31.5 in / 80 cm 15 mil Code 128 1.34 in / 3.4 cm 25.6 in / 65 cm 10 mil DM 2.84 in / 7.2 cm 11.7 in / 29.7 cm 6,7 mil PDF417 3.3 in / 8.4 cm 9.6 in / 24.4 cm 15 mil QR Code 1.3 in / 3.3 cm 15.35 in / 39 cm
Communication	
Host Interface supported	USB / RS232
Mechanical	
Dimension	87.8mm x 177.7mm x 71.6 mm (L x H X W)
Weight	147g
Trigger Switch Life	10 million times

Functionality

Symbologies	1D: Codabar, Code 11, Code 128, Code 2 of 5, Code 39, Code 93 and 93i, EAN/JAN-13, EAN/JAN 8, IATA Code 2 of 5, Interleaved 2 of 5, Matrix 2 of 5, MSI, GS1 Databar, UPC-A, UPC E, UPC-A/EAN-13 with Extended Coupon Code, Coupon GS1 Code 32(PARAF), EAN-UCC Emulation, GS1 Data Bar
	2D: 2D Stacked: Codablock A, Codablock F, PDF417, MicroPDF417
	2D Matrix: Aztec Code, Data Matrix, MaxiCode, QR Code, Chinese Sensible (Han Xin), Grid Matrix, Dot Code
	Postal Codes: Australian Post, British Post, Canadian Post, China Post, Japanese Post, Korea Post, Netherlands Post, Planet Code, Postnet
Electrical	
Operation Voltage	4 VDC to 5.5 VDC
Current Consumption	Operation mode: 252mA, Standby mode: 77mA
Indicator	Buzzer, LED
Environmental	
ESD Protection	Functional after 8K Contact and 15K Air
Mechanical Shock	2.1M onto concrete (scanner only)
Operating Temperature	-10°C to 50°C (14°F to 122°F)
Storage Temperature	-40°C to 70°C (-40°F to 158°F)
Relative Humidity	95% non-condensing
IP Rating	IP42
Regulatory Approvals	
CE, FCC, BSMI, VCCI	

1.4 Getting Started

To get started with MS852Plus, please connect USB cable to the USB port of a host PC.



1.5 LED Indicator / Beeper Sequence

Description	Indication	
	Beeper	LED
Trigger Pull	No Sound	No Light
No decode		
Wake up		
Decode	Middle Tone	Green Blink
Snapshot started	Low Tone	Blinking
Snapshot is complete	Low Tone	No Light
Power on	Low Tone, Middle Tone, High Tone	No Light
Transmission error	Four Low Tones	No Light
Entry error	Low Tone, High Tone	Green Blink
Defaults set	High Tone, Low Tone, High Tone, Low Tone	Green Blink
Parameter entered		
Number entry expected	High Tone, Low Tone	Green Blink

Chapter 2 – Command Settings

2.1 General setting

2.1.1 Default

Scan below bar code to restore the factory setting.

Default



2.1.2 Display F/W Version

Scan below bar code to display F/W version.

Display F/W Version



2.2 USB Device Type

Scan the following bar code to program the scan engine.

USB HID Keyboard*



USB Serial



Note: No extra configuration (e.g., baud rate) is necessary.

2.3 Scan mode

2.3.1 Scan Mode

Manual Trigger-Normal*



Normal mode offers good scan speed and the longest working ranges (depth of field).

Presentation Mode



Presentation Mode uses ambient light and scan engine illumination to detect bar codes. When in Presentation Mode, the LEDs remain dim until a bar code is presented to the scan engine, then the aimer turns on and the LEDs turn up to read the code. If the light level in the room is not high enough, Presentation Mode may not work properly. Scan the following bar code to program your scan engine for Presentation Mode.

2.3.2 Aimer Mode

This feature allows you to turn the aimer on and off. When the Interlaced bar code is scanned, the aimer is interlaced with the illumination LEDs.

Default = Interlaced.

Aimer Mode*



Off



2.3.3 Mobile Phone Read Mode

When this mode is selected, your scan engine is optimized to read bar codes from mobile phone or other LED displays. However, the speed of scanning printed bar codes may be slightly lower when this mode is enabled.

You can enable Mobile Phone Reading for either a hand held device, or for a hands-free (presentation) application.

Hand Held Scanning - Mobile Phone



Streaming Presentation - Mobile Phone



2.4 Suffix Selection

2.4.1 Add CR Suffix All Symbolologies

Scan the following bar code if you wish to add a carriage return suffix to all symbologies at once.

Add CR Suffix All Symbolologies



2.4.2 Suffix Selections

Add Suffix



Clear All Suffixes



2.5 LED and Beeper

2.5.1 LED on Good Decode

The LED indicator can be programmed On or Off in response to a good read.
Default = On.

LED-Good Read On*



LED-Good Read Off



2.5.2 Beep After Good Read

The beeper may be programmed On or Off in response to a good read.
Turning this option off only turns off the beeper response to a good read indication. All error and menu beeps are still audible.
Default = Beeper - Good Read On.

Beeper-Good Read ON*



Beeper-Good Read Off



2.5.3 Beep After Good Read

Low Volume



Medium Volume



High Volume *



2.5.4 Beep Tone

Low Tone



Medium Tone *



High Tone



2.6 USB Country Keyboard Types

If your interface is USB Keyboard or Keyboard Wedge, your keyboard layout default is a US keyboard. To change this layout, refer to the chart below for your keyboard country. Scan the appropriate bar code below to change the layout.

United States*



Albania



Azeri (Cyrillic)



Azeri (Latin)



Belarus



Belgium



Brazil



Brazil (MS)



Bosnia



Bulgaria (Cyrillic)



Bulgaria (Latin)



Canada (French)



Canada (French legacy)



Canada (Multilingual)



Croatia



Czech



Czech (Programmers)



Czech (QWERTY)



Czech (QWERTZ)



Denmark



Dutch (Netherlands)



Faroese



France



Germany



Greek (220 Latin)



Greek (319 Latin)



Estonia



Finland



Gaelic



Greek



Greek (220)



Greek (319)



Greek (Latin)



Greek (Polytonic)



Hungarian (101 key)



Iceland



Italian (142)



Japan ASCII



Italy



Irish



Hungary



Hebrew



Greek (MS)



Kyrgyz (Cyrillic)



Kazakh



Latvia



Latin America



Lithuania



Latvia (QWERTY)



Lithuania (IBM)



Macedonia



Malta



Mongolian (Cyrillic)



Norway



Poland



Polish (214)



Polish (Programmers)



Portugal



Romania



Russia



Russian (MS)



Russian (Typewriter)



SCS



Serbia (Cyrillic)



Serbia (Latin)



Slovakia (QWERTY)



Slovakia



Slovenia



Slovakia (QWERTZ)



Slovenia



Spain



Spanish variation



Sweden



Switzerland (French)



Switzerland (German)



Tatar



Turkey F



Turkey Q



Ukrainian



United Kingdom



United States (Dvorak)



United States (Dvorak left)



United States (Dvorak right)



United States (International)



Uzbek (Cyrillic)



Chapter 3 – Symbology

3.1 All symbologies

If you want to decode all the symbologies allowable for your scanner, scan the All Symbologies On code. If on the other hand, you want to decode only a particular symbology, scan All Symbologies Off followed by the On symbol for that particular symbology.

Note: Scanner performance may reduce by scanning All Symbologies On.

Only scan All Symbologies On when needed.

When All Symbologies On is scanned, 2D Postal Codes are not enabled.

2D Postal Codes must be enabled separately.

All Symbologies On



All Symbologies Off



3.2 Message Length Description

You are able to set the valid reading length of some of the bar code symbologies. You may wish to set the same value for mini-mum and maximum length to force the scanner to read fixed length bar code data. This helps reduce the chances of a misread.

EXAMPLE: Decode only those bar codes with a count of 9-20 characters.

Min.length = 09 Max. length = 20

EXAMPLE: Decode only those bar codes with a count of 15 characters.

Min. length = 15 Max. length = 15

For a value other than the minimum and maximum message length defaults, scan the bar codes included in the explanation of the symbology, then scan the digit value of the message length and Save bar codes on the [Programming Chart](#). The minimum and maximum lengths and the defaults are included with the respective symbologies.

3.3 1D

3.3.1 Codabar

Codabar On*



Default All Codabar Settings



Codabar Off



3.3.1.1 Codabar Start / Stop Characters

Start/Stop characters identify the leading and trailing ends of the bar code. You may either transmit, or not transmit Start/Stop characters.

Default = Don't Transmit.

Transmit



Don't Transmit*



3.3.1.2 Codabar Check Characters

Codabar check characters are created using different “modulos.” You can program the scanner to read only Codabar bar codes with Modulo 16 check characters. Default = No Check Character.

No Check Character indicates that the scanner reads and transmits bar code data with or without a check character.

When Check Character is set to **Validate and Transmit**, the scanner will only read Codabar bar codes printed with a check character, and will transmit this character at the end of the scanned data.

When Check Character is set to **Validate, but Don't Transmit**, the unit will only read Codabar bar codes printed with a check character, but will not transmit the check character with the scanned data.

No Check Character*



Validate Modulo 16, but Don't Transmit



Validate Modulo 16 and Transmit



3.3.1.3 Codabar Concatenation

Codabar supports symbol concatenation. When you enable concatenation, the scanner looks for a Codabar symbol having a “D” start character, adjacent to a symbol having a “D” stop character. In this case the two messages are concatenated into one with the “D” characters omitted.



Select Require to prevent the scanner from decoding a single “D” Codabar symbol without its companion. This selection has no effect on Codabar symbols without Stop/Start D characters.

On



Off*



Require



3.3.1.4 Codabar Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information.

Minimum and Maximum lengths = 2-60.

Minimum Default = 4, Maximum Default = 60.

Minimum Message Length



Maximum Message Length



3.3.2 Code 39

Code 39 On*



Default All Code 39 Settings



Code 39 Off



3.3.2.1 Code 39 Start / Stop Characters

Start/Stop characters identify the leading and trailing ends of the bar code. You may either transmit, or not transmit Start/ Stop characters.

Default = Don't Transmit.

Transmit



Don't Transmit*



3.3.2.2 Code 39 Check Characters

No Check Character indicates that the scanner reads and transmits bar code data with or without a check character.

When Check Character is set to **Validate, but Don't Transmit**, the unit only reads Code 39 bar codes printed with a check character, but will not transmit the check character with the scanned data.

When Check Character is set to **Validate and Transmit**, the scanner only reads Code 39 bar codes printed with a check character, and will transmit this character at the end of the scanned data. Default = No Check Character.

No Check Character*



Validate , but Don't Transmit



Validate and Transmit



3.3.2.3 Code 39 Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information.

Minimum and Maximum lengths = 0-48.

Minimum Default = 0, Maximum Default = 48.

Minimum Message Length



Maximum Message Length



3.3.2.4 Code 39 Append

This function allows the scanner to append the data from several Code 39 bar codes together before transmitting them to the host computer. When the scanner encounters a Code 39 bar code with the append trigger character(s), it buffers Code 39 bar codes until it reads a Code 39 bar code that does not have the append trigger. The data is then transmitted in the order in which the bar codes were read (FIFO). Default = Off.

Off *



On



3.3.2.5 Code 32 Pharmaceutical (PARAF)

Code 32 Pharmaceutical is a form of the Code 39 symbology used by Italian pharmacies. This symbology is also known as PARAF.

Note: Trioptic Code must be turned off while scanning Code 32 Pharmaceutical codes.

Off *



On



3.3.2.6 Full ASCII

Full ASCII Code 39 decoding is enabled, certain character pairs within the bar code symbol will be interpreted as a single character.

For example: \$V will be decoded as the ASCII character SYN, and /C will be decoded as the ASCII character #. Default = Off.

NUL %U	DLE \$P	SP SPACE	0 0	@ %V	P P	' %W	p +P
SOH \$A	DC1 \$Q	! /A	1 1	A A	Q Q	a +A	q +Q
STX \$B	DC2 \$R	" /B	2 2	B B	R R	b +B	r +R
ETX \$C	DC3 \$S	# /C	3 3	C C	S S	c +C	s +S
EOT \$D	DC4 \$T	\$ /D	4 4	D D	T T	d +D	t +T
ENQ \$E	NAK \$U	% /E	5 5	E E	U U	e +E	u +U
ACK \$F	SYN \$V	& /F	6 6	F F	V V	f +F	v +V
BEL \$G	ETB \$W	' /G	7 7	G G	W W	g +G	w +W
BS \$H	CAN \$X	(/H	8 8	H H	X X	h +H	x +X
HT \$I	EM \$Y	) /I	9 9	I I	Y Y	i +I	y +Y
LF \$J	SUB \$Z	* /J	:	/Z	J J	Z Z	j +J z +Z
VT \$K	ESC %A	+ /K	; %F	K K	[%K	k +K	{ %P
FF \$L	FS %B	, /L	< %G	L L	\ %L	l +L	%Q
CR \$M	GS %C	- -	= %H	M M	] %M	m +M	} %R
SO \$N	RS %D	. .	> %I	N N	^ %N	n +N	~ %S
SI \$O	US %E	/ /O	? %J	O O	_ %O	o +O	DEL %T

Character pairs /M and /N decode as a minus sign and period respectively.
Character pairs /P through /Y decode as 0 through 9.

Full ASCII Off *



Full ASCII On



3.3.2.7 Code 39 Code Page

Code pages define the mapping of character codes to characters. If the data received does not display with the proper characters, it may be because the bar code being scanned was created using a code page that is different from the one the host program is expecting. If this is the case, scan the bar code below, select the code page with which the bar codes were created ([see ISO 2022/ISO 646 Character Replacements](#)), and scan the value and the Save bar code from the [Programming Chart](#). The data characters should then appear properly.

Code 39 Code Page



3.3.3 Interleaved 2 of 5

Interleaved 2 of 5 On*



Default All Interleaved 2 of 5 Settings



Interleaved 2 of 5 Off



3.3.3.1 Check digit

No Check Digit indicates that the scanner reads and transmits bar code data with or without a check digit. When Check Digit is set to Validate, but Don't Transmit, the unit only reads Interleaved 2 of 5 bar codes printed with a check digit, but will not transmit the check digit with the scanned data. When Check Digit is set to Validate and Transmit, the scanner only reads Interleaved 2 of 5 bar codes printed with a check digit, and will transmit this digit at the end of the scanned data. Default = No Check Digit.

No Check Digit*



Validate , but Don't Transmit



Validate and Transmit



3.3.3.2 Interleaved 2 of 5 Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information.

Minimum and Maximum lengths = 2-80.

Minimum Default = 4, Maximum Default = 80.

Minimum Message Length



Maximum Message Length



3.3.4 Code 93

Code 93 On*



Default All Code 93 Settings



Code 93 Off



3.3.4.1 Code 93 Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 0-80. Minimum Default = 0, Maximum Default = 80.

Minimum Message Length



Maximum Message Length



3.3.4.2 Code 93 Append

This function allows the scanner to append the data from several Code 93 bar codes together before transmitting them to the host computer. When this function is enabled, the scanner stores those Code 93 bar codes that start with a space (excluding the start and stop symbols), and does not immediately transmit the data. The scanner stores the data in the order in which the bar codes are read, deleting the first space from each. The scanner transmits the appended data when it reads a Code 93 bar code that starts with a character other than a space. Default = Off.

Off *



On



3.3.4.3 Code 93 Code Page

Code pages define the mapping of character codes to characters. If the data received does not display with the proper characters, it may be because the bar code being scanned was created using a code page that is different from the one the host program is expecting. If this is the case, scan the bar code below, select the code page with which the bar codes were created ([see ISO 2022/ISO 646 Character Replacements](#)), and scan the value and the Save bar code from the [Programming Chart](#). The data characters should then appear properly.

Code 93 Code Page



3.3.5 Straight 2 of 5 Industrial (three-bar start/stop)

Straight 2 of 5
Industrial On



Default All Straight 2 of 5
Industrial Settings



Straight 2 of 5
Industrial Off*



3.3.5.1 Straight 2 of 5 Industrial Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-48. Minimum Default = 4, Maximum Default = 48.

Minimum Message Length



Maximum Message Length



3.3.6 Straight 2 of 5 IATA (two-bar start/stop)

Straight 2 of 5 IATA

Industrial On



Default All Straight 2 of 5
IATA Industrial Settings



Straight 2 of 5 IATA

Industrial Off*



3.3.6.1 Straight 2 of 5 IATA Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-48. Minimum Default = 4, Maximum Default = 48.

Minimum Message Length



Maximum Message Length



3.3.7 Matrix 2 of 5

Matrix 2 of 5 On



Default All Matrix 2 of 5
Settings



Matrix 2 of 5 Off*



3.3.7.1 Matrix 2 of 5 Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-80. Minimum Default = 4, Maximum Default = 80.

Minimum Message Length



Maximum Message Length



3.3.8 Code 11

Code 11 On



Default All Code 11 Settings



Code 11 Off*



3.3.8.1 Check Digits Required

This option sets whether 1 or 2 check digits are required with Code 11 bar codes. Default = Two Check Digits.

One Check Digit



Two Check Digits*



3.3.8.2 Code 11 Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-80. Minimum Default = 4, Maximum Default = 80.

Minimum Message Length



Maximum Message Length



3.3.9 Code 128

Code 128 On*



Default All Code 128
Settings



Code 128 Off



3.3.9.1 ISBT 128 Concatenation

In 1994 the International Society of Blood Transfusion (ISBT) ratified a standard for communicating critical blood information in a uniform manner. The use of ISBT formats requires a paid license. The ISBT 128 Application Specification describes 1) the critical data elements for labeling blood products, 2) the current recommendation to use Code 128 due to its high degree of security and its space-efficient design, 3) a variation of Code 128 that supports concatenation of neighboring symbols, and 4) the standard layout for bar codes on a blood product label. Use the bar codes below to turn concatenation on or off. Default =Off.

On



Off*



3.3.9.2 Code 128 Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 0-80. Minimum Default = 0, Maximum Default = 80.

Minimum Message Length



Maximum Message Length



3.3.9.3 Code 128 Append

This function allows the scanner to append the data from several Code 128 bar codes together before transmitting them to the host computer. When the scanner encounters a Code 128 bar code with the append trigger character(s), it buffers Code 128 bar codes until it reads a Code 128 bar code that does not have the append trigger. The data is then transmitted in the order in which the bar codes were read (FIFO). Default = On.

On*



Off



3.3.9.4 Code 128 Code Page

Code pages define the mapping of character codes to characters. If the data received does not display with the proper characters, it may be because the bar code being scanned was created using a code page that is different from the one the host program is expecting. If this is the case, scan the bar code below, select the code page with which the bar codes were created ([see ISO 2022/ISO 646 Character Replacements](#)), and scan the value and the Save bar code from the [Programming Chart](#). The data characters should then appear properly.

Code 128 Code Page



3.3.10 UPC-A

Code UPC-A On*



Default All UPC-A Settings



Code UPC-A Off



Note: To convert UPC-A bar codes to EAN-13, see [Convert UPC-A to EAN-13](#).

3.3.10.1 UPC-A Check Digit

This selection allows you to specify whether the check digit should be transmitted at the end of the scanned data or not. Default = On.

On*



Off



3.3.10.2 UPC-A Number System

The numeric system digit of a U.P.C. symbol is normally transmitted at the beginning of the scanned data, but the unit can be programmed so it will not transmit it. Default = On.

On*



Off



3.3.10.3 UPC-A Addenda

This selection adds 2 or 5 digits to the end of all scanned UPC-A data. Default = Off for both 2 Digit and 5 Digit Addenda.

2 Digit Addenda Off*



2 Digit Addenda On



5 Digit Addenda Off*



5 Digit Addenda On



3.3.10.4 UPC-A Addenda Required

When Required is scanned, the scanner will only read UPC-A bar codes that have addenda. You must then turn on a 2 or 5 digit addenda. Default = Not Required.

Not Required*



Required



3.3.10.5 Addenda Timeout

You can set a time during which the scanner looks for an addenda. If an addenda is not found within this time period, the data can be either transmitted or discarded, based on the setting you are using for UPC-A Addenda Required. Set the length (in milliseconds) for this timeout by scanning the bar code below, then setting the timeout (from 0-65535 milliseconds) by scanning digits from the [Programming Chart](#), then scanning Save. Default = 100.

Addenda Timeout



3.3.10.6 UPC-A Addenda Separator

When this feature is on, there is a space between the data from the bar code and the data from the addenda. When turned off, there is no space. Default = On.

On *



Off



3.3.10.7 UPC-A/EAN-13 with Extended Coupon Code

Use the following codes to enable or disable UPC-A and EAN-13 with Extended Coupon Code. When left on the default setting (Off), the scanner treats Coupon Codes and Extended Coupon Codes as single bar codes. If you scan the Allow Concatenation code, when the scanner sees the coupon code and the extended coupon code in a single scan, it transmits both as one symbology. Otherwise, it transmits the first coupon code it reads. If you scan the Require Concatenation code, the scanner must see and read the coupon code and extended coupon code in a single read to transmit the data. No data is output unless both codes are read. Default = Off.

Off*



Allow Concatenation



Require Concatenation



3.3.10.8 Coupon GS1 DataBar Output

If you scan coupons that have both UPC and GS1 DataBar codes, you may wish to scan and output only the data from the GS1 DataBar code. Scan the GS1 Output On code below to scan and output only the GS1 DataBar code data. Default = GS1 Out-put Off.

GS1 Output Off *



GS1 Output On



3.3.11 UPC-E0

Most U.P.C. bar codes lead with the 0 number system. To read these codes, use the UPC-E0 On selection. If you need to read codes that lead with the 1 number system, use [UPC-E1](#). Default = On.

Code UPC-A On*



Default All UPC-E Settings



Code UPC-A Off



3.3.11.1 UPC-E0 Expand

UPC-E Expand expands the UPC-E code to the 12 digit, UPC-A format. Default = Off.

Off *



On



3.3.11.2 UPC-E0 Addenda Required

When Required is scanned, the scanner will only read UPC-E bar codes that have addenda. Default = Not Required.

Not Required *



Required



3.3.11.3 UPC-E0 Addenda Separator

When this feature is On, there is a space between the data from the bar code and the data from the addenda. When turned Off, there is no space.

Default = On.

On*



Off



3.3.11.4 UPC-E0 Check Digit

Check Digit specifies whether the check digit should be transmitted at the end of the scanned data or not. Default = On.

On*



Off



3.3.11.5 UPC-E0 Leading Zero

This feature allows the transmission of a leading zero (0) at the beginning of scanned data. To prevent transmission, scan Off. Default = On.

On*



Off



3.3.11.6 UPC-E0 Addenda

This selection adds 2 or 5 digits to the end of all scanned UPC-E data.

Default = Off for both 2 Digit Addenda and 5 Digit Addenda.

2 Digit Addenda Off*



2 Digit Addenda On



5 Digit Addenda Off*



5 Digit Addenda On



3.3.12 UPC-E1

Most U.P.C. bar codes lead with the 0 number system. For these codes, use [UPC-E0](#). If you need to read codes that lead with the 1 number system, use the UPC-E1 On selection. Default = Off.

Off*



On



3.3.13 EAN/JAN-13

EAN/JAN-13 On*



Default All EAN/JAN
Settings



EAN/JAN-13 Off



3.3.13.1 Convert UPC-A to EAN-13

When UPC-A Converted to EAN-13 is selected, UPC-A bar codes are converted to 13 digit EAN-13 codes by adding a zero to the front. When Do not Convert UPC-A is selected, UPC-A codes are read as UPC-A.

Do not Convert UPC-A*



UPC-A Converted to EAN-13



3.3.13.2 EAN/JAN-13 Check Digit

This selection allows you to specify whether the check digit should be transmitted at the end of the scanned data or not. Default = On.

On*



Off



3.3.13.3 EAN/JAN-13 Addenda

This selection adds 2 or 5 digits to the end of all scanned EAN/JAN-13 data.
Default = Off for both 2 Digit Addenda and 5 Digit Addenda.

2 Digit Addenda Off*



2 Digit Addenda On



5 Digit Addenda Off*



5 Digit Addenda On



3.3.13.4 EAN/JAN-13 Addenda Required

When Required is scanned, the scanner will only read EAN/JAN-13 bar codes that have addenda. Default = Not Required.

Not Required*



Required



3.3.13.5 EAN/JAN-13 Addenda Separator

When this feature is **On**, there is a space between the data from the bar code and the data from the addenda. When turned **Off**, there is no space.

Default = On.

Note: If you want to enable or disable EAN13 with Extended Coupon Code, refer to

[UPC-A/EAN-13 with Extended Coupon Code.](#)

On*



Off



3.3.13.6 ISBN Translate

When On is scanned, EAN-13 Bookland symbols are translated into their equivalent ISBN number format. Default = Off.

On



Off *



3.3.14 EAN/JAN-8

EAN/JAN-8 On*



Default All EAN/JAN-8
Settings

EAN/JAN-8 Off



3.3.14.1 EAN/JAN-8 Check Digit

This selection allows you to specify whether the check digit should be transmitted at the end of the scanned data or not. Default = On.

On*



Off



3.3.14.2 EAN/JAN-8 Addenda

This selection adds 2 or 5 digits to the end of all scanned EAN/JAN-8 data.
Default = Off for both 2 Digit and 5 Digit Addenda.

2 Digit Addenda Off*



2 Digit Addenda On



5 Digit Addenda Off*



5 Digit Addenda On



3.3.14.3 EAN/JAN-8 Addenda Required

When Required is scanned, the scanner will only read EAN/JAN-8 bar codes that have addenda. Default = Not Required.

Not Required*



Required



3.3.14.4 EAN/JAN-8 Addenda Separator

When this feature is On, there is a space between the data from the bar code and the data from the addenda. When turned Off, there is no space.

Default = On.

On*



Off



3.3.15 MSI

MSI On



Default All MSI Settings



MSI Off *



3.3.15.1 MSI Check Character

Different types of check characters are used with MSI bar codes. You can program the scanner to read MSI bar codes with Type 10 check characters. Default = Validate Type 10, but Don't Transmit. When Check Character is set to Validate Type 10/11 and Transmit, the scanner will only read MSI bar codes printed with the specified type check character(s), and will transmit the character(s) at the end of the scanned data. When Check Character is set to Validate Type 10/11, but Don't Transmit, the unit will only read MSI bar codes printed with the specified type check character(s), but will not transmit the check character(s) with the scanned data.

Validate Type 10, but Don't Transmit*



Validate Type 10 and Transmit



Validate 2 Type 10 Characters, but Don't Transmit



Validate 2 Type 10 Characters and Transmit



Validate Type 10 then Type 11 Character, but Don't Transmit



Validate Type 10 then Type 11 Character and Transmit



Disable MSI Check Characters



3.3.15.2 MSI Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 4-48. Minimum Default = 4, Maximum Default = 48.

Minimum Message Length



Maximum Message Length



3.3.16 NEC 2 of 5

NEC 2 of 5 On



Default All NEC 2 of 5 Settings



NEC 2 of 5 Off *



3.3.16.1 Check Digit

No Check Digit indicates that the scanner reads and transmits bar code data with or without a check digit. When Check Digit is set to Validate, but Don't Transmit, the unit only reads NEC 2 of 5 bar codes printed with a check digit, but will not transmit the check digit with the scanned data. When Check Digit is set to Validate and Transmit, the scanner only reads NEC 2 of 5 bar codes printed with a check digit, and will transmit this digit at the end of the scanned data. Default = No Check Digit.

No Check Digit*



Validate and Transmit



Validate, but Don't Transmit



3.3.16.2 NEC 2 of 5 Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 2-80 Minimum Default = 4, Maximum Default = 80

Minimum Message Length



Maximum Message Length



3.3.17 GS1-128

GS1-128 On



Default All GS1-128 Settings

GS1-128 Off *



3.3.17.1 GS1-128 Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-80 Minimum Default = 1, Maximum Default = 80

Minimum Message Length



Maximum Message Length



3.3.18 Telepen

Telepen On



Default All Telepen Settings

Telepen Off *



3.3.18.1 Telepen Output

Using AIM Telepen Output, the scanner reads symbols with start/stop pattern 1 and decodes them as standard full ASCII (start/stop pattern 1). When Original Telepen Output is selected, the scanner reads symbols with start/stop pattern 1 and decodes them as compressed numeric with optional full ASCII (start/stop pattern 2). Default = AIM Telepen Output.

AIM Telepen Output *



Original Telepen Output



3.3.18.2 Telepen Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-60 Minimum Default = 1, Maximum Default = 60

Minimum Message Length



Maximum Message Length



3.4 2D

3.4.1 GS1 DataBar Omnidirectional

GS1 DataBar
Omnidirectional On*



Default All GS1 DataBar
Omnidirectional Settings



GS1 DataBar
Omnidirectional Off



3.4.2 GS1 DataBar Limited

GS1 DataBar Limited On*



Default All GS1 DataBar
Limited Settings



GS1 DataBar Limited Off



3.4.3 GS1 DataBar Expanded

GS1 DataBar Expanded On*



Default All GS1 DataBar
Expanded Settings



GS1 DataBar Expanded Off



3.4.3.1 GS1 DataBar Expanded Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 4-74. Minimum Default = 4, Maximum Default = 74.

Minimum Message Length



Maximum Message Length



3.4.4 PDF 417

PDF 417 On*



Default All PDF 417 Settings



PDF 417 Off



3.4.4.1 PDF417 Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-2750. Minimum Default = 1, Maximum Default = 2750.

Minimum Message Length



Maximum Message Length



3.4.5 MacroPDF417

MacroPDF417 is an implementation of PDF417 capable of encoding very large amounts of data into multiple PDF417 bar codes. When this selection is enabled, these multiple bar codes are assembled into a single data string. Default = On.

On *



Off



3.4.6 MicroPDF417

MicroPDF417 On



Default All MicroPDF417
Settings



MicroPDF417 Off *



3.4.6.1 MicroPDF417 Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-366. Minimum Default = 1, Maximum Default = 366.

Minimum Message Length



Maximum Message Length



3.4.7 GS1 Composite Codes

Linear codes are combined with a unique 2D composite component to form a new class called GS1 Composite symbology. GS1 Composite symbologies allow for the co-existence of symbologies already in use. Default = Off.

On



Off *



3.4.7.1 UPC/EAN Version

Scan the UPC/EAN Version On bar code to decode GS1 Composite symbols that have a U.P.C. or an EAN linear component. (This does not affect GS1 Composite symbols with a GS1-128 or GS1 linear component.)

Default = UPC/EAN Version Off.

Note: If you scan coupons that have both UPC and GS1 DataBar codes, you may wish to scan and output only the data from the GS1 DataBar code. See Coupon GS1 DataBar Output for further information.

On



Off *



3.4.7.2 GS1 Composite Code Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-2435. Minimum Default = 1, Maximum Default = 2435.

Minimum Message Length



Maximum Message Length



3.4.7.3 GS1 Emulation

The scanner can automatically format the output from any GS1 data carrier to emulate what would be encoded in an equivalent GS1-128 or GS1 DataBar symbol. GS1 data carriers include UPC-A and UPC-E, EAN-13 and EAN-8, ITF-14, GS1-128, and GS1-128 DataBar and GS1 Composites. (Any application that accepts GS1 data can be simplified since it only needs to recognize one data carrier type.)

If GS1-128 Emulation is scanned, all retail codes (U.P.C., UPC-E, EAN8, EAN13) are expanded out to 16 digits. If the AIM ID is enabled, the value will be the GS1-128 AIM ID,]C1.

If GS1 DataBar Emulation is scanned, all retail codes (U.P.C., UPC-E, EAN8, EAN13) are expanded out to 16 digits. If the AIM ID is enabled, the value will be the GS1-DataBar AIM ID,]em.

If GS1 Code Expansion Off is scanned, retail code expansion is disabled, and UPC-E expansion is controlled by the UPC-E0 Expand setting.

If the AIM ID is enabled, the value will be the GS1-128 AIM ID,]C1.

If EAN8 to EAN13 Conversion is scanned, all EAN8 bar codes are converted to EAN13 format. Default = GS1 Emulation Off.

GS1-128 Emulation



GS1 Emulation Off *



GS1 Code Expansion Off



GS1 DataBar Emulation



EAN8 to EAN13 Conversion



3.4.8 TCIF Linked Code 39 (TLC39)

This code is a composite code since it has a Code 39 linear component and a MicroPDF417 stacked code component. All bar code readers are capable of reading the Code 39 linear component. The MicroPDF417 component can only be decoded if TLC39 On is selected. The linear component may be decoded as Code 39 even if TLC39 is off. Default = Off.

On



Off *



3.4.9 QR Code

QR Code On*



Default All QR Code Settings



QR Code Off



3.4.9.1 QR Code Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-7089. Minimum Default = 1, Maximum Default = 7089.

Minimum Message Length



Maximum Message Length



3.4.9.2 QR Code Append

This function allows the scanner to append the data from several QR Code bar codes together before transmitting them to the host computer. When the scanner encounters an QR Code bar code with the append trigger character(s), it buffers the number of QR Code bar codes determined by information encoded in those bar codes. Once the proper number of codes is reached, the data is output in the order specified in the bar codes. Default = On.

On*



Off



3.4.9.3 QR Code Page

QR Code pages define the mapping of character codes to characters. If the data received does not display with the proper characters, it may be because the bar code being scanned was created using a code page that is different from the one the host program is expecting. If this is the case, scan the bar code below, select the code page with which the bar codes were created ([see ISO 2022/ISO 646 Character Replacements](#)), and scan the value and the Save bar code from the [Programming Chart](#). The data characters should then appear properly.

QR Code Page



3.4.10 Data Matrix

Data Matrix On*



Default All Data Matrix
Settings



Data Matrix Off



3.4.10.1 Data Matrix Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-3116. Minimum Default = 1, Maximum Default = 3116.

Minimum Message Length



Maximum Message Length



3.4.10.2 Data Matrix Append

This function allows the scanner to append the data from several Data Matrix bar codes together before transmitting them to the host computer. When the scanner encounters an Data Matrix bar code with the append trigger character(s), it buffers the number of Data Matrix bar codes determined by information encoded in those bar codes. Once the proper number of codes is reached, the data is output in the order specified in the bar codes. Default = On.

On*



Off



3.4.10.3 Data Matrix Code Page

Data Matrix Code pages define the mapping of character codes to characters. If the data received does not display with the proper characters, it may be because the bar code being scanned was created using a code page that is different from the one the host program is expecting. If this is the case, scan the bar code below, select the code page with which the bar codes were created ([see ISO 2022/ISO 646 Character Replacements](#)), and scan the value and the Save bar code from the [Programming Chart](#). The data characters should then appear properly.

Data Matrix Code Page



3.4.11 MaxiCode

MaxiCode On



Default All MaxiCode Settings



MaxiCode Off*



3.4.11.1 MaxiCode Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-150. Minimum Default = 1, Maximum Default = 150..

Minimum Message Length



Maximum Message Length



3.4.12 Chinese Sensible (Han Xin) Code

Han Xin On



Default All Han Xin Settings



Han Xin Off*



3.4.12.1 Han Xin Code Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-7833. Minimum Default = 1, Maximum Default = 7833.

Minimum Message Length



Maximum Message Length



3.4.13 Postal Codes - 2D

The following lists the possible 2D postal codes, and 2D postal code combinations that are allowed. Only one 2D postal code selection can be active at a time. If you scan a second 2D postal code selection, the first selection is overwritten. Default = 2D Postal Codes Off.

2D Postal Codes Off *



3.4.13.1 Single 2D Postal Codes

Australian Post On



British Post On



Canadian Post On



Intelligent Mail Bar Code On



Japanese Post On



KIX Post On



Planet Code On



Postal-4i On



Postnet On



Postnet with B and B' Fields On



InfoMail On



3.4.13.2 Combination 2D Postal Codes

InfoMail and British Post On



**Postal-4i and Postnet with B
and B' Fields On**



Postnet and Postal-4i On



Planet Code and Postal-4i On



**Postal-4i and Intelligent Mail Bar
Code On**



**Planet Code, Postnet, and
Postal-4i On**



Planet Code and Postnet On



**Planet Code, Postal-4i, and
Intelligent Mail Bar Code On**



**Intelligent Mail Bar Code and
Postnet with B and B' Fields On**



**Planet Code, Postal-4i, and
Postnet with B and B' Fields On**



**Postnet and Intelligent Mail Bar
Code On**



**Postal-4i, Intelligent Mail Bar
Code, and Postnet with B and
B' Fields On**



**Planet Code and Postnet with B
and B' Fields On**



**Planet Code, Postal-4i,
Intelligent Mail Bar Code, and
Postnet with B and B' Fields On**



**Planet Code and Intelligent Mail
Bar Code On**



**Planet Code, Postal-4i,
Intelligent Mail Bar Code, and
Postnet On**



**Planet Code, Postnet, and
Intelligent Mail Bar Code On**



**Postnet, Postal-4i, and
Intelligent Mail Bar Code On**



**Planet Code, Intelligent Mail Bar
Code, and Postnet with B and B'
Fields On**



3.4.13.3 Planet Code Check Digit

This selection allows you to specify whether the check digit should be transmitted at the end of Planet Code data. Default = Don't Transmit.

Transmit Check Digit



Don't Transmit Check Digit*



3.4.13.4 Postnet Check Digit

This selection allows you to specify whether the check digit should be transmitted at the end of Postnet data. Default = Don't Transmit.

Transmit Check Digit



Don't Transmit Check Digit *



3.4.13.5 Australian Post Interpretation

This option controls what interpretation is applied to customer fields in Australian 4-State symbols. Bar Output lists the bar patterns in "0123" format. Numeric N Table causes that field to be interpreted as numeric data using the N Table. Alphanumeric C Table causes the field to be interpreted as alphanumeric data using the C Table. Refer to the Australian Post Specification Tables. Combination C and N Tables causes the field to be interpreted using either the C or N Tables.

Bar Output *



Numeric N Table



Alphanumeric C Table



Combination C and N Tables



3.4.14 Postal Codes – Linear

China Post (Hong Kong 2 of 5)

China Post (Hong Kong 2 of 5) On



Default All China Post (Hong Kong 2 of 5) Settings

China Post (Hong Kong 2 of 5) Off*



3.4.14.1 China Post (Hong Kong 2 of 5) Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 2-80. Minimum Default = 4, Maximum Default = 80.

Minimum Message Length



Maximum Message Length



3.4.15 Trioptic Code

Note: If you are going to scan Code 32 Pharmaceutical codes, Trioptic Code must be off. Trioptic Code is used for labeling magnetic storage media.

On



Off *



3.4.16 Codablock A

Codablock A On



Default All Codablock A
Settings



Codablock A Off*



3.4.16.1 Codablock A Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-600. Minimum Default = 1, Maximum Default = 600.

Minimum Message Length



Maximum Message Length



3.4.17 Codablock F

Codablock F On



Default All Codablock F
Settings



Codablock F Off*



3.4.17.1 Codablock F Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-2048. Minimum Default = 1, Maximum Default = 2048.

Minimum Message Length



Maximum Message Length



3.4.17.2 Label Code

The standard Label Code is used in libraries. Default = Off.

On



Off *



3.4.18 Aztec Code

Aztec Code On*



Default All Aztec Code
Settings



Aztec Code Off



3.4.18.1 Aztec Code Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-3832. Minimum Default = 1, Maximum Default = 3832.

Minimum Message Length



Maximum Message Length



3.4.18.2 Aztec Append

This function allows the scanner to append the data from several Aztec bar codes together before transmitting them to the host computer. When the scanner encounters an Aztec bar code with the append trigger character(s), it buffers the number of Aztec bar codes determined by information encoded in those bar codes. Once the proper number of codes is reached, the data is output in the order specified in the bar codes. Default = On.

On *



Off



3.4.18.3 Aztec Code Page

Aztec Code pages define the mapping of character codes to characters. If the data received does not display with the proper characters, it may be because the bar code being scanned was created using a code page that is different from the one the host program is expecting. If this is the case, scan the bar code below, select the code page with which the barcodes were created ([see ISO 2022/ISO 646 Character Replacements](#)), and scan the value and the Save bar code from the [Programming Chart](#).

The data characters should then appear properly.

Aztec Code Page



3.4.19 Chinese Sensible (Han Xin) Code

Han Xin On



Default All Han Xin Settings



Han Xin Off *



3.4.19.1 Han Xin Code Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 1-7833. Minimum Default = 1, Maximum Default = 7833.

Minimum Message Length



Maximum Message Length



3.4.20 Korea Post

Korea Post On



Default All Korea Post
Settings



Korea Post Off *



3.4.20.1 Korea Post Message Length

Scan the bar codes below to change the message length. Refer to [Message Length Description](#) for additional information. Minimum and Maximum lengths = 2-80. Minimum Default = 4, Maximum Default = 48.

Minimum Message Length



Maximum Message Length



3.4.20.2 Korea Post Check Digit

This selection allows you to specify whether the check digit should be transmitted at the end of the scanned data. Default = Don't Transmit.

Transmit Check Digit



Don't Transmit Check Digit *



Appendix A – Symbology Charts

“m” represents the AIM modifier character. Refer to International Technical Specification, Symbology Identifiers, for AIM modifier character details.

Prefix/Suffix entries for specific symbologies override the universal (All Symbologies, 99) entry.

Refer to Data Editing and Data Formatting for information about using Code ID and AIM ID.

Linear Symbologies

Symbology	AIM		Honeywell	
	ID	Possible modifiers (m)	ID	Hex
All Symbologies				99
Codabar	]Fm	0-1	a	61
Code 11	]H3		h	68
Code 128	]Cm	0, 1, 2, 4	j	6A
Code 32 Pharmaceutical (PARAF)	]X0		<	3C
Code 39 (supports Full ASCII mode)	]Am	0, 1, 3, 4, 5, 7	b	62
TCIF Linked Code 39 (TLC39)	]L2		T	54
Code 93 and 93i	]Gm	0-9, A-Z, a-m	i	69
EAN	]Em	0, 1, 3, 4	d	64
EAN-13 (including Bookland EAN)	]E0		d	64
EAN-13 with Add-On	]E3		d	64
EAN-13 with Extended Coupon Code	]E3		d	64
EAN-8	]E4		D	44
EAN-8 with Add-On	]E3		D	44
GS1				
GS1 DataBar	]em	0	y	79
GS1 DataBar Limited	]em		{	7B
GS1 DataBar Expanded	]em		}	7D
GS1-128	]C1		l	49
2 of 5				
China Post (Hong Kong 2 of 5)	]X0		Q	51
Interleaved 2 of 5	]Im	0, 1, 3	e	65
Matrix 2 of 5	]X0		m	6D
NEC 2 of 5	]X0		Y	59
Straight 2 of 5 IATA	]Rm	0, 1, 3	f	66
Straight 2 of 5 Industrial	]S0		f	66
MSI	]Mm	0, 1	g	67
Telepen	]Bm		t	74
UPC		0, 1, 2, 3, 8, 9, A, B, C		

Symbology	AIM		Honeywell	
	ID	Possible modifiers (m)	ID	Hex
UPC-A	JE0		c	63
UPC-A with Add-On	JE3		c	63
UPC-A with Extended Coupon Code	JE3		c	63
UPC-E	JE0		E	45
UPC-E with Add-On	JE3		E	45
UPC-E1	JX0		E	45

Add Honeywell Code ID				5C80
Add AIM Code ID				5C81
Add Backslash				5C5C
Batch mode quantity			5	35

2D Symbologies

Symbology	AIM		Honeywell	
	ID	Possible modifiers (m)	ID	Hex
All Symbologies				99
Aztec Code	]zm	0-9, A-C	z	7A
Chinese Sensible Code (Han Xin Code)	]X0		H	48
Codablock A	]O6	0, 1, 4, 5, 6	V	56
Codablock F	]Om	0, 1, 4, 5, 6	q	71
Code 49	]Tm	0, 1, 2, 4	l	6C
Data Matrix	]dm	0-6	w	77
GS1	]em	0-3	y	79
GS1 Composite	]em	0-3	y	79
GS1 DataBar Omnidirectional	]em		y	79
MaxiCode	]Um	0-3	x	78
PDF417	]Lm	0-2	r	72
MicroPDF417	]Lm	0-5	R	52
QR Code	]Qm	0-6	s	73
Micro QR Code	]Qm		s	73

Postal Symbologies

Symbology	AIM		Honeywell	
	ID	Possible modifiers (m)	ID	Hex
All Symbologies				99
Australian Post	]X0		A	41
British Post	]X0		B	42
Canadian Post	]X0		C	43
China Post	]X0		Q	51
InfoMail	]X0		,	2c
Intelligent Mail Bar Code	]X0		M	4D
Japanese Post	]X0		J	4A
KIX (Netherlands) Post	]X0		K	4B
Korea Post	]X0		?	3F
Planet Code	]X0		L	4C
Postal-4i	]X0		N	4E
Postnet	]X0		P	50

ASCII Conversion Chart (Code Page 1252)

In keyboard applications, ASCII Control Characters can be represented in 3 different ways, as shown below. The CTRL+X function is OS and application dependent. The following table lists some commonly used Microsoft functionality. This table applies to U.S. style keyboards. Certain characters may differ depending on your Country Code/PC regional settings.

Non-printable ASCII control characters			Keyboard Control + ASCII (CTRL+X) Mode		
			Windows Mode Control + X Mode On (KBDCAS2)		
DEC	HEX	Char	Control + X Mode Off (KBDCAS0)	CTRL + X	CTRL + X function
0	00	NUL	Reserved	CTRL+ @	
1	01	SOH	NP Enter	CTRL+ A	Select all
2	02	STX	Caps Lock	CTRL+ B	Bold
3	03	ETX	ALT Make	CTRL+ C	Copy
4	04	EOT	ALT Break	CTRL+ D	Bookmark
5	05	ENQ	CTRL Make	CTRL+ E	Center
6	06	ACK	CTRL Break	CTRL+ F	Find
7	07	BEL	Enter / Ret	CTRL+ G	
8	08	BS	(Apple Make)	CTRL+ H	History
9	09	HT	Tab	CTRL+ I	Italic
10	0A	LF	(Apple Break)	CTRL+ J	Justify
11	0B	VT	Tab	CTRL+ K	hyperlink
12	0C	FF	Delete	CTRL+ L	list, left align
13	0D	CR	Enter / Ret	CTRL+ M	
14	0E	SO	Insert	CTRL+ N	New
15	0F	SI	ESC	CTRL+ O	Open
16	10	DLE	F11	CTRL+ P	Print
17	11	DC1	Home	CTRL+ Q	Quit
18	12	DC2	PrtScn	CTRL+ R	
19	13	DC3	Backspace	CTRL+ S	Save
20	14	DC4	Back Tab	CTRL+ T	
21	15	NAK	F12	CTRL+ U	
22	16	SYN	F1	CTRL+ V	Paste
23	17	ETB	F2	CTRL+ W	
24	18	CAN	F3	CTRL+ X	
25	19	EM	F4	CTRL+ Y	
26	1A	SUB	F5	CTRL+ Z	
27	1B	ESC	F6	CTRL+ [	
28	1C	FS	F7	CTRL+ \	
29	1D	GS	F8	CTRL+]	
30	1E	RS	F9	CTRL+ ^	
31	1F	US	F10	CTRL+ -	
127	7F	Δ	NP Enter		

ISO 2022/ISO 646 Character Replacements

Code pages define the mapping of character codes to characters. If the data received does not display with the proper characters, it may be because the bar code being scanned was created using a code page that is different from the one the host pro-gram is expecting. If this is the case, select the code page with which the bar codes were created. The data characters should then appear properly.

Code Page Selection Method/Country	Standard	Keyboard Country	Honeywell Code Page Option
United States (standard ASCII)	ISO/IEC 646-IRV	n/a	1
Automatic National Character Replacement	ISO/IEC 2022	n/a	2 (default)
Binary Code page	n/a	n/a	3
<i>Default "Automatic National Character replacement" will select the below Honeywell Code Page options for Code128, Code 39 and Code 93.</i>			
United States	ISO/IEC 646-06	0	1
Canada	ISO /IEC 646-121	54	95
Canada	ISO /IEC 646-122	18	96
Japan	ISO/IEC 646-14	28	98
China	ISO/IEC 646-57	92	99
Great Britain (UK)	ISO /IEC 646-04	7	87
France	ISO /IEC 646-69	3	83
Germany	ISO/IEC646-21	4	84
Switzerland	ISO /IEC 646-CH	6	86
Sweden / Finland (extended Annex C)	ISO/IEC 646-11	2	82
Ireland	ISO /IEC 646-207	73	97
Denmark	ISO/IEC 646-08	8	88
Norway	ISO/IEC 646-60	9	94
Italy	ISO/IEC 646-15	5	85
Portugal	ISO/IEC 646-16	13	92
Spain	ISO/IEC 646-17	10	90
Spain	ISO/IEC 646-85	51	91

Dec			35	36	64	91	92	93	94	96	123	124	125	126
Hex			23	24	40	5B	5C	5D	5E	60	7B	7C	7D	7E
US	0	1	#	\$	@	[	\	]	^	`	{		}	~
CA	54	95	#	\$	à	â	ç	é	î	ô	é	ù	è	û
CA	18	96	#	\$	à	â	ç	é	É	ô	é	ù	è	û
JP	28	98	#	\$	@	[	¥	]	^	`	{		}	~
CN	92	99	#	¥	@	[	\	]	^	`	{		}	~
GB	7	87	E	\$	@	[	\	]	^	`	{		}	~
FR	3	83	E	\$	à	°	ç	§	^	µ	é	ù	è	~
DE	4	84	#	\$	§	Ä	Ö	U	^	`	ä	ö	ü	ß
CH	6	86	ù	\$	à	é	ç	é	î	ô	ä	ö	ü	û
SE/FI	2	82	#	H	É	Ä	Ö	Ä	U	é	ä	ö	ä	ü
DK	8	88	#	\$	@	Æ	Ø	Å	^	`	æ	ø	å	~
NO	9	94	#	\$	@	Æ	Ø	Å	^	`	æ	ø	å	~
IE	73	97	E	\$	Ó	É	Í	Ú	Á	ó	é	í	ú	á
IT	5	85	E	\$	§	°	ç	é	^	ù	ä	ö	è	ì
PT	13	92	#	\$	§	Ä	Ç	Ö	^	`	ä	ç	ö	°
ES	10	90	#	\$	§	í	Ñ	¿	^	`	°	ñ	ç	~
ES	51	91	#	\$	·	í	Ñ	Ç	¿	`	·	ñ	ç	~
COUNTRY	Country Keyboard	Honeywell CodePage	ISO / IEC 646 National Character Replacements											

Appendix B – Programming Chart

0



1



2



3



4



5



6



7



8



9



A



B



C



D



E



F



Save



Reset



Discard



Note: If you make an error while scanning the letters or digits (before scanning Save), scan Discard, scan the correct letters or digits, and **Save** again.